

Thoriso Mbambisa
Johannesburg, Gauteng

Email: thorisombambisa004@gmail.com
LinkedIn: [linkedin.com/in/thoriso-mbambisa-613988269](https://www.linkedin.com/in/thoriso-mbambisa-613988269)
GitHub Portfolio: github.com/Thoriso04
Virtual Portfolio: <https://thoriso04.github.io/Virtual-CV/>
Tel: +(27) 61 214 1260

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL, HTML5, CSS, C#
Frameworks & Libraries: React, Node.js, .NET, TailwindCSS
Databases: PostgreSQL, MySQL, MongoDB, Microsoft SQL Server
Tools & Platforms: Git, GitHub, Docker, Azure, VS Code, Kafka, MATLAB
Methodologies: Agile/Scrum, CI/CD, RESTful API Design, Waterfall

EDUCATION

BSc Honours in Computer Science and Information Technology: North-West University (NWU) *January 2026 – Present* | GPA 64.25 %

- Relevant Modules: Artificial Intelligence, Data Warehouse, Databases, Information Systems Engineering
- Activities: Object-Oriented Programming student assistant and tutor.

BSc Information Technology: North-West University (NWU) *February 2023 – December 2025* | GPA: 67.34 %

- Relevant Modules: Data Structures & Algorithms, Object-Oriented Programming, Databases, Systems Analysis and Design, User Interface Programming, Operating Systems, Computer Networks, Data Analytics, Decision Support Systems, Artificial Intelligence, Information Security
- Activities: Basic computer literacy tutor work and student assistant

Matric Certificate (Bachelor's Pass): Boksburg High School
January 2022 – December 2022 | GPA: 66.86

PROJECTS

Real-Time E-Commerce Streaming Pipeline: Cloud Data Engineering Project

Tech Stack: Python, Azure Event Hubs, Azure Cosmos DB, Azure Stream Analytics, Azure SQL Database, Power BI

- Built a cloud-native streaming data pipeline that ingests 400+ events per second from simulated ecommerce sources, processes them through real-time SQL queries with windowing and anomaly detection, and stores results across two databases simultaneously.
- Implemented a Power BI dashboard with AI-powered Key Influencers visuals, conditional alerting, and DirectQuery live refresh to surface actionable business insights, including cart abandonment prioritisation and customer behaviour patterns.
- Designed fault-tolerant architecture with dead-letter routing, duplicate detection, and late-arrival handling across Azure's cloud platform, demonstrating production-grade stream processing principles.

NWU Classroom Polling System: Full Stack Web Development Project

Tech Stack: REACT 19, Node.js, TypeScript, PostgreSQL, Redis, Socket.io, Docker, Nginx

- **Developed** a real-time, high-concurrency engagement platform using Socket.IO for sub-millisecond question delivery and live response synchronization, supporting up to 15,000 lines of code across a 14week development cycle.
- **Engineered** a secure, multi-role architecture (Student, Lecturer, Admin) featuring JWT authentication, bcrypt hashing, and Redis-backed caching to optimize performance and session management.
- **Implemented** comprehensive data analytics and reporting modules, allowing lecturers to export live participation statistics and aggregated results in CSV and PDF formats while ensuring POPIAcompliant anonymous guest access.
- **Designed** a fully responsive, cross-device UI and a multi-layer containerized environment using Docker to ensure seamless scalability and consistent deployment across university infrastructure.

CERTIFICATIONS & CONTINUOUS LEARNING

- MATLAB Onramp (2026)
- MATLAB Data Analysis (2026)
- Artificial Intelligence and Deep Learning Onramp with MATLAB (2026)

LEADERSHIP & ACTIVITIES

- First-Year Leadership Camp at North-West University (2023)
- Served as class representative for MTHS 113, facilitating lecturer-student communication
- NWU "Leave a Legacy" Program Participant (2023)
- Deputy Head Boy (2022)